The Artificial Neuron

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: adds compares multiplies

1. Should I go? Answer 1 (yes) or 0 (no). Then choose how much each one matters to you (0 to 3).

	Answer (1 or 0)	**How much it matters**	**Answer $\times$ how much**
Is it sunny?			
Are my friends going?			
Is my homework done?			
		Total	

Is the total 4 or more? _____ Do I go? _____ Did my partner decide the same? _____

2. Multiply, add, compare.

An artificial neuron _____ each input by its weight, _____ them up, and _____ the total with the threshold.

Inputs: 1, 0, 1. Weights: 3, 2, -1. Threshold: 2.

Total = $(1 \times 3) + (0 \times 2) + (1 \times -1)$ = _____ Does it fire (2 or more)? _____

Inputs: 1, 1, 1. Weights: 1, -2, -1. Threshold: 2.

Total = _____ Does it fire? _____

3. Be a neuron. Weights: **A:** +2, **B:** +2, **C:** -1. Threshold: **3 or more**.

Round	**Inputs that are on**	**Total**	**Fire?**
1	A and B		
2	A and C		
3	A, B, and C		
4	B and C		
5	A only		

4. Real neuron, artificial neuron. Match each part.

Real neuron	**Artificial neuron**
Dendrites take signals in	
Strong or weak synapses	
Fires only if the total is big enough	

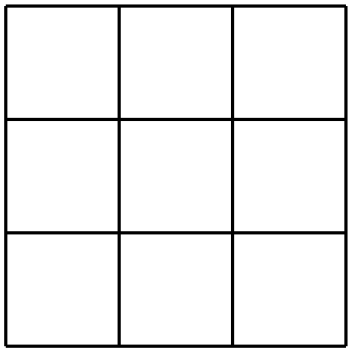
5. Talk about it. ChatGPT has billions of weights. Nobody can choose them by hand.

I think an AI with billions of weights gets them by _____

_____

_____

Go Further (Day 2). A neuron that sees. Write the weights that got the most test pictures right.



Threshold: _____

Test pictures right: _____ out of 6

Which squares got **positive** weights? Why? _____

Which squares got **negative** weights? Why? _____